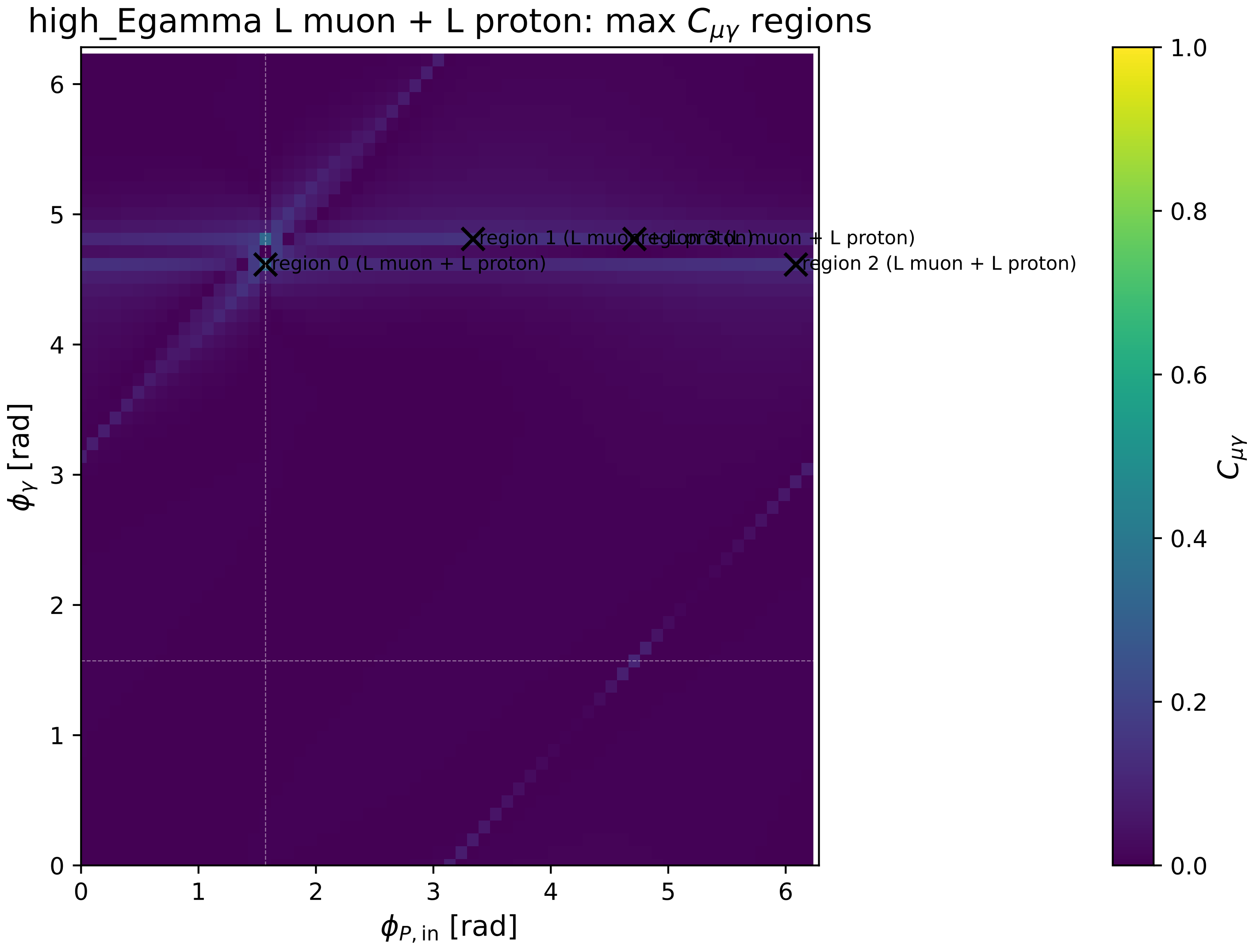
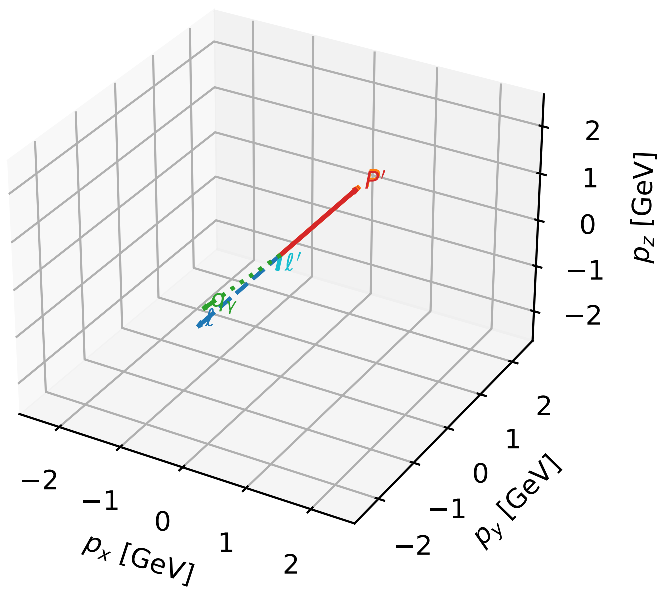

high_Egamma L muon + L proton: max $C_{\mu\gamma}$ regions
region 0 (L muon + L proton)
region 1 (L muon + L proton)
region 2 (L muon + L proton)
region 3 (L muon + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L muon + L proton characteristic max $C_{\mu\gamma}$ configuration

3D momenta

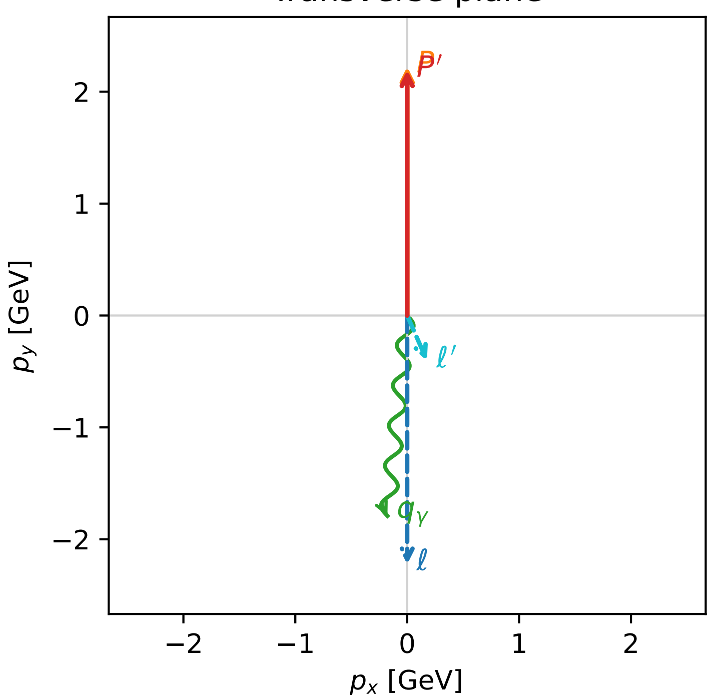



```
c_e_gamma_high_egamma_ll_region_0  C_{mu\gamma}=0.339661
region: high_Egamma_LL_region_0  final-pair delta_xy=0.510997 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
```
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

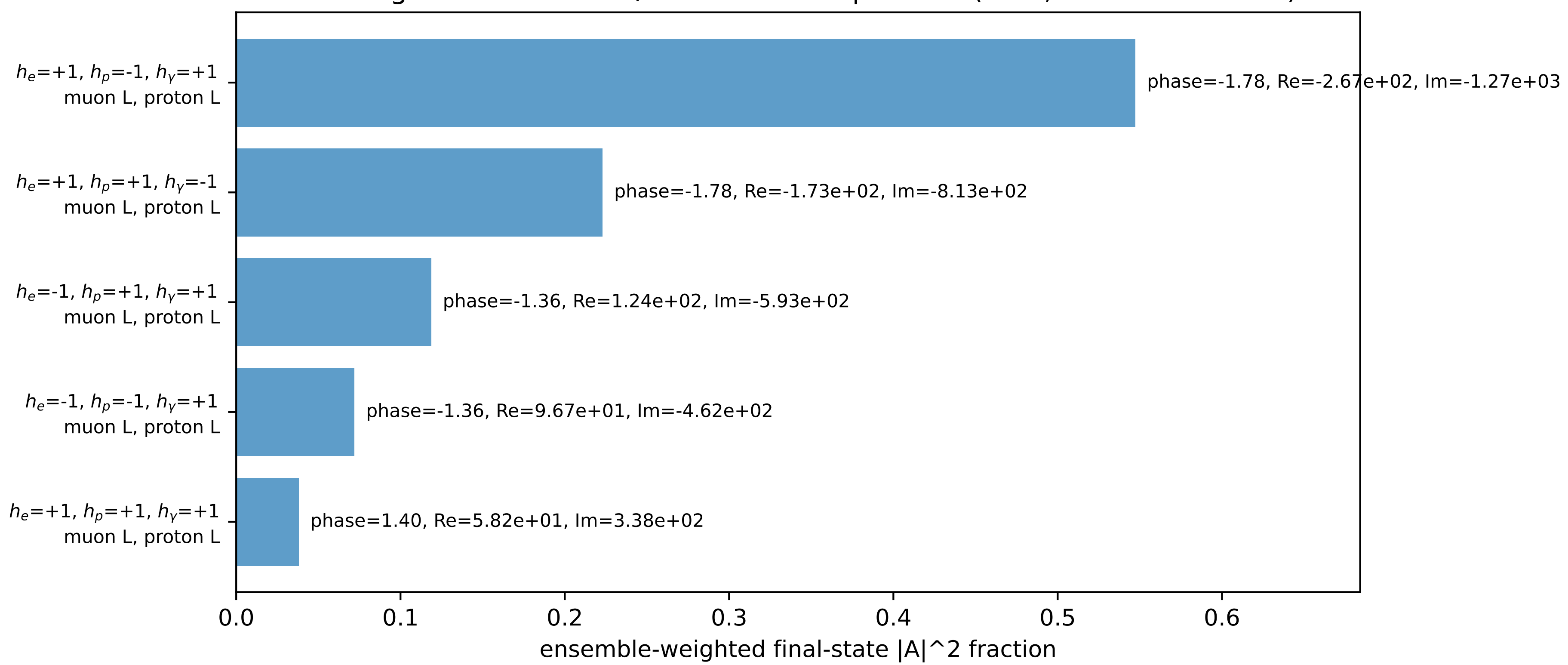
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.19415
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1.8, ϕ_γ=4.61421
Q^2=0.199845, x_B=0.0120018, t=-0.000128163, W^2=17.3312, y=0.807338
$\theta(\ell',\gamma)$=29.278 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 0.4523 p_x= 0.1764 p_y= -0.4028 p_z= 0.0000
P' E= 2.3862 p_x= 0.0000 p_y= 2.1942 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.1764 p_y= -1.7913 p_z= 0.0000

Transverse plane

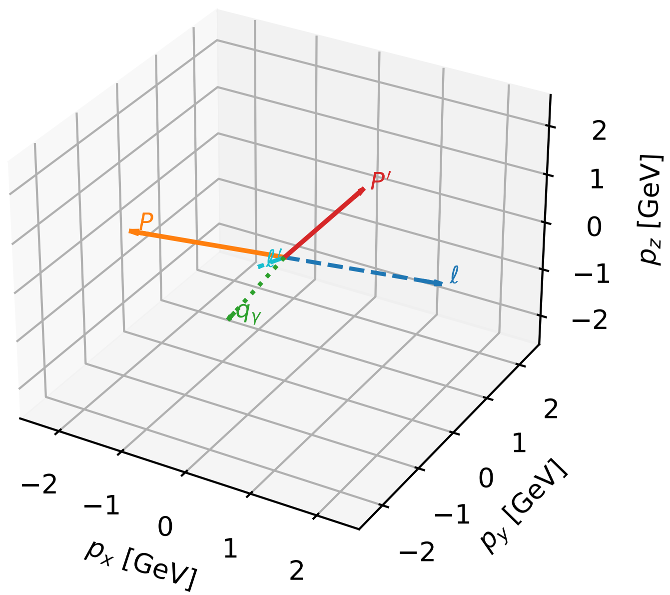


Leading initial-ensemble/final-state components (N=5, retained=99.9%)



L muon + L proton characteristic max $C_{\mu\gamma}$ configuration

3D momenta

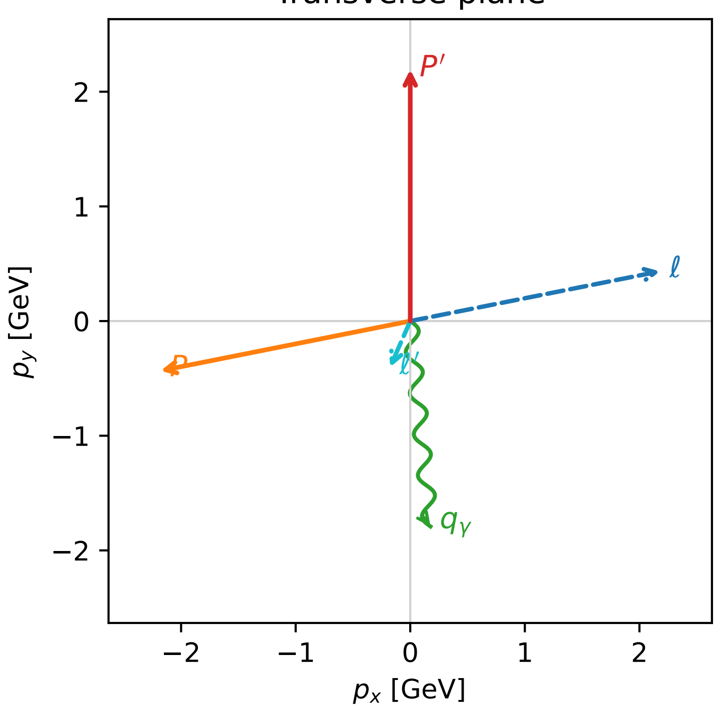


c_e_gamma_high_egamma_ll_region_1 C_{mu\gamma}=0.14156
region: high_Egamma_LL_region_1 final-pair delta_xy=0.510997 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

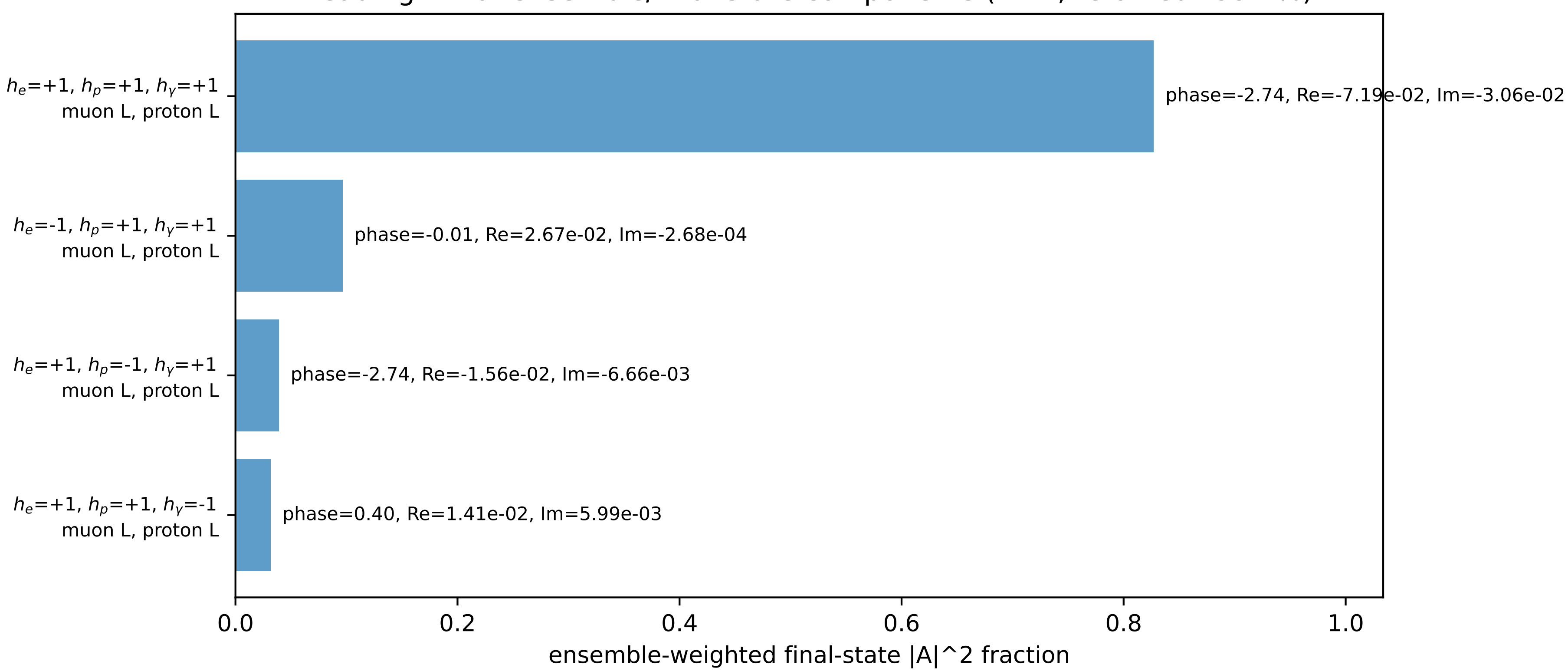
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.19415
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.19635, ϕ_P=3.33794
E_γ=1.8, ϕ_γ=4.81056
Q^2=3.10966, x_B=0.158972, t=-11.6591, W^2=17.3312, y=0.948421
$\theta(\ell', \gamma)$=29.278 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= -2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 0.4523 p_x= -0.1764 p_y= -0.4028 p_z= 0.0000
P' E= 2.3862 p_x= 0.0000 p_y= 2.1942 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.1764 p_y= -1.7913 p_z= 0.0000

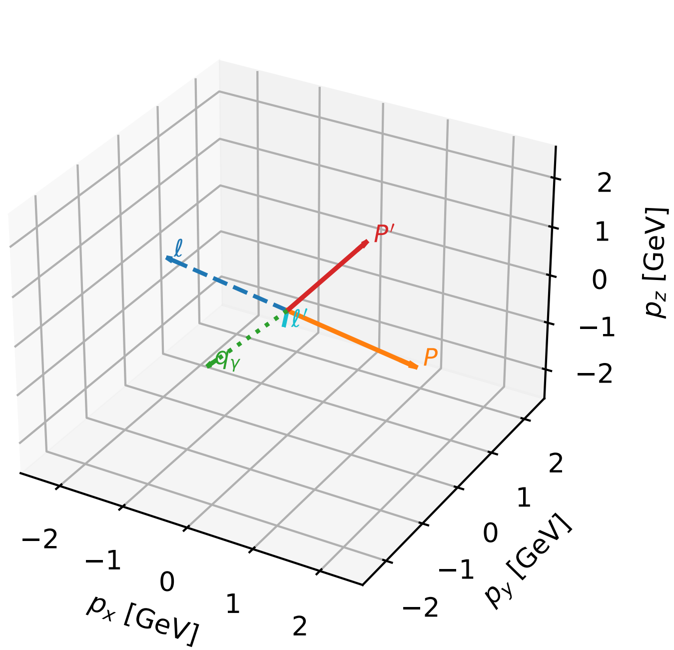
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.4%)



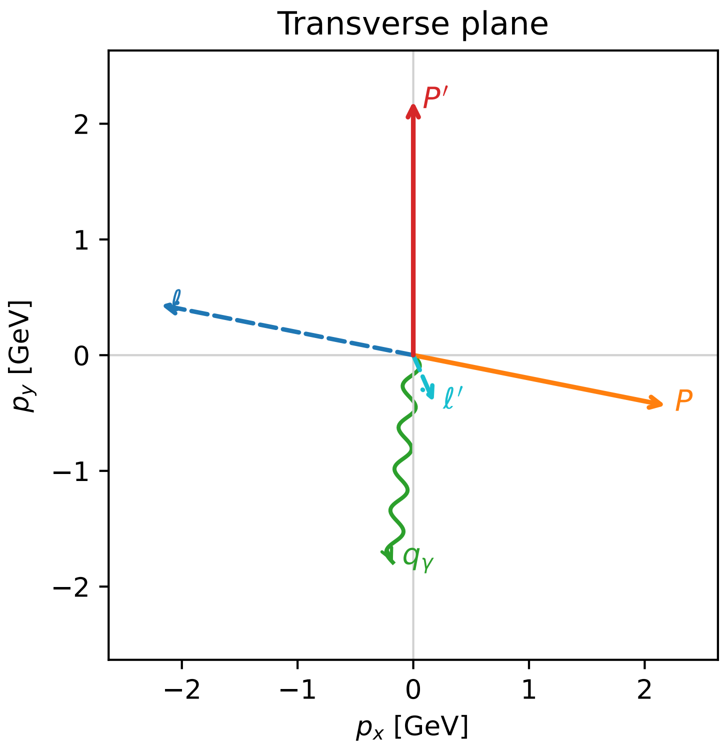
L muon + L proton characteristic max $C_{\mu\gamma}$ configuration

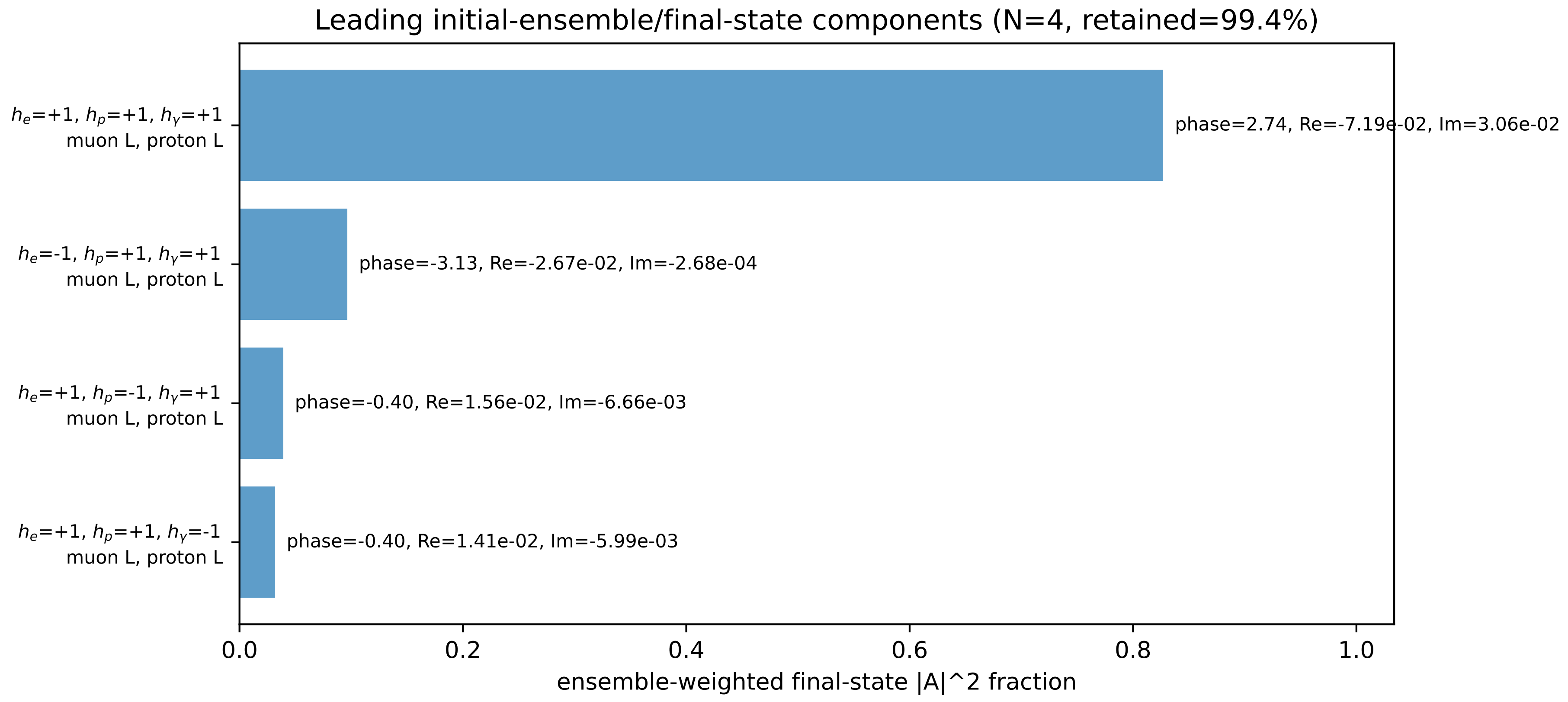


c_e_gamma_high_egamma_ll_region_2 C_{mu\gamma}=0.14156
region: high_Egamma_LL_region_2 final-pair delta_xy=0.510997 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.19415
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.94524, ϕ_P=6.08684
E_γ=1.8, ϕ_γ=4.61421
Q^2=3.10966, x_B=0.158972, t=-11.6591, W^2=17.3312, y=0.948421
$\theta(\ell',\gamma)$=29.278 deg

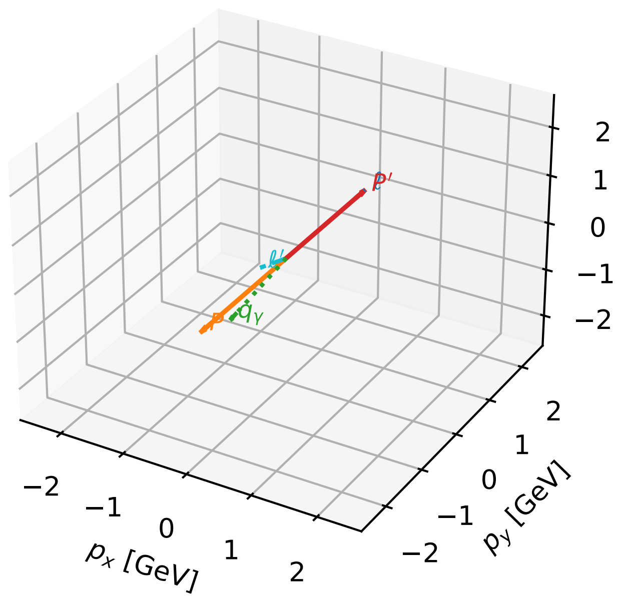
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= 2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 0.4523 p_x= 0.1764 p_y= -0.4028 p_z= 0.0000
P' E= 2.3862 p_x= 0.0000 p_y= 2.1942 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.1764 p_y= -1.7913 p_z= 0.0000





L muon + L proton characteristic max $C_{\mu\gamma}$ configuration

3D momenta

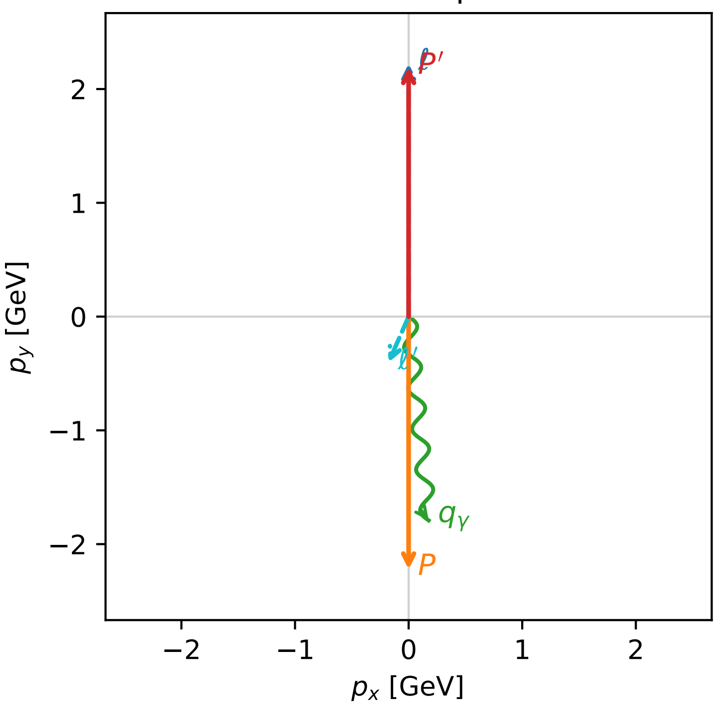



```
c_e_gamma_high_egamma_ll_region_3  C_{mu\gamma}=0.125109
region: high_Egamma_LL_region_3  final-pair delta_xy=0.510997 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
```
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.19415
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=4.81056
Q^2=3.78189, x_B=0.186914, t=-19.5115, W^2=17.3312, y=0.981014
$\theta(\ell', \gamma)$=29.278 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 0.4523 p_x= -0.1764 p_y= -0.4028 p_z= 0.0000
P' E= 2.3862 p_x= 0.0000 p_y= 2.1942 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.1764 p_y= -1.7913 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=100.0%)